Session 5 · Deck 5.B3

Many Dimensions, and Finite Elements

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Where We Are

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S1 ✓	S2 ✓	S3 ✓	S4 ✓	S5	S6	S7	S8	S9	S10
Sep 24	Oct 1	Oct 8	Oct 15	Oct 22	Oct 29	Nov 5	Nov 12	Nov 19	Dec 3
Intro	HA · OLG	Num. DP	Numerics	Numerics	HA classic	ML	RL	Text	Agents
Growth	Policy	Python	I	II	The wall	Deep nets	HA + DL	LLMs	Projects
trunk	branches	the machine · classical methods			the wall	new tools		new data	colleagues

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Block B — projection: 5.B1 the residual · 5.B2 choosing a basis · **5.B3** many dimensions and finite elements · 5.B4 coefficients and the worked example.

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Roadmap: Four Parts

5.B2 chose a basis on an interval. Most economic problems have several state variables, and some have kinks. This deck handles both.

Part	Topic	The question it answers
1	Tensor products and the curse	How does a one-dimensional basis become a d-dimensional one, and what does it cost?
2	Two ways out	Which terms can be dropped without losing much?
3	Finite elements	What does a *local* basis look like, and why does a kink stop being a problem?
4	Choosing	Global and sparse, or local and dense: which fails where?

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By the end: you will know why a two-state model is cheap and a ten-state one is not, what Smolyak and complete polynomials throw away, and how a tent basis turns a borrowing limit into a node.

The Tensor Product, and the Curse of Dimensionality

Multiply one-dimensional polynomials together, one factor per state variable. Orthogonality survives; the count of terms explodes.

Chebyshev polynomials are defined on $[-1, 1]$; most problems in economics are multidimensional. To approximate $F : [-1, 1]^d \to \mathbb{R}$ at degree κ, take the **tensor product**:

$$\hat{F}(x) = \sum_{n_1=0}^{\kappa} \cdots \sum_{n_d=0}^{\kappa} \theta_{n_1 \ldots n_d} \underbrace{T_{n_1}(x_1) \cdots T_{n_d}(x_d)}_{\text{one factor per state}}.$$

Key property: if the one-dimensional basis is orthogonal under a weight, the tensor-product basis is orthogonal under the product weight. Everything in 5.B2 survives.

The cost: each index runs over $\kappa + 1$ values and there are d of them, so the basis has $(\kappa + 1)^d$ terms, up to $x_1^\kappa x_2^\kappa \cdots x_d^\kappa$ — exponential in d.

d	terms, $\kappa = 4$
2	25
3	125
5	3,125
10	9,765,625

Three ways out

Sparse grids (Smolyak, 4.B1); low-rank tensor decompositions; adaptive basis selection (S7). Part 2 takes the first, after a simpler idea.

Ledger `chebyshev.tensor_vs_complete`. **The same curse 3.A2 met with grids, arriving through a different door.**

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The First Fix: Complete Polynomials

Cap the total degree instead of each variable's degree. The high-order interactions go; for a smooth function they were the smallest terms.

The tensor basis includes $x_1^{\kappa} \cdots x_d^{\kappa}$, a term of total degree $d\kappa$. Gaspar & Judd (1997) keep only the terms whose exponents **sum** to at most κ:

$$P_{\kappa}^{d} \equiv \left\{ x_1^{i_1} \cdots x_d^{i_d} \;\middle|\; \sum_{j=1}^{d} i_j \leq \kappa, \;\; i_j \geq 0 \right\}.$$

Idea

Instead of letting each variable reach degree κ independently, restrict the sum of the exponents. Only low-complexity interactions survive.

Advantage: the count falls from $(\kappa+1)^d$ to $\binom{\kappa+d}{d}$.
Disadvantage: still large for high d and κ.

d	tensor	complete
2	25	15
3	125	35
5	3,125	126
10	9,765,625	**1,001**

$\kappa = 4$

$\binom{\kappa+d}{d}$ grows like a polynomial in d for fixed κ, not an exponential. At ten states the basis is a thousand terms, not ten million.

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Gaspar & Judd (1997), "Solving Large-Scale Rational-Expectations Models," *Macroeconomic Dynamics* 1(1), 45–75.

Which Terms Survive: $d = 2$, $\kappa = 2$

Two variables, degree two. Sort the nine tensor terms by total degree; the complete basis keeps the first three rows.

i_1+i_2	term	kept by
0	1	both
1	$x_1,\ x_2$	both
2	$x_1^2,\ x_1x_2,\ x_2^2$	both
3	$x_1^2x_2,\ x_1x_2^2$	tensor only
4	$x_1^2x_2^2$	tensor only

The tensor rule caps each index separately and keeps all $3 \times 3 = \mathbf{9}$. The complete rule caps the *sum* and keeps $\mathbf{6}$.

The rule in one line

Keep $x_1^{i_1} \cdots x_d^{i_d}$ if $\sum_j i_j \leq \kappa$. Throw away everything else.

Same rule at $d = 3$, $\kappa = 2$: tensor $3^3 = 27$ terms, complete $\binom{5}{3} = \mathbf{10}$. The gap widens with every dimension added. What is lost is the high-order interactions — for a smooth function the smallest terms, which is why the trade is usually free.

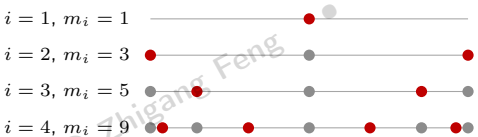
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Ledger `chebyshev.tensor_vs_complete`.

Smolyak, Recalled: Chebyshev Extrema, Level by Level

4.B1 built Smolyak's sparse grid from four symbols. Here they are once more, and the nodes they sit on.

- d: the number of **state variables**, one direction each.
- **Level** i: how finely one direction is resolved. Level i has m_i nodes: $m_1 = 1$ (the centre), $m_i = 2^{i-1} + 1$ after that $(3, 5, 9, 17)$, roughly doubling.
- μ: **one budget of refinement steps, shared by all directions**. Raising direction j to level i_j spends $i_j - 1$ steps: $\sum_j (i_j - 1) \le \mu$.
- q: the same budget counted in levels, $\sum_j i_j \le d + \mu \equiv q$.



red: new at this level; grey: kept from the level below.

Built on Chebyshev? Yes — the extrema

The nodes of level i are the **extrema** of $T_{m_i - 1}$:

$$x_j = \cos\Big(\frac{\pi j}{m_i - 1}\Big), \quad j = 0, \dots, m_i - 1.$$

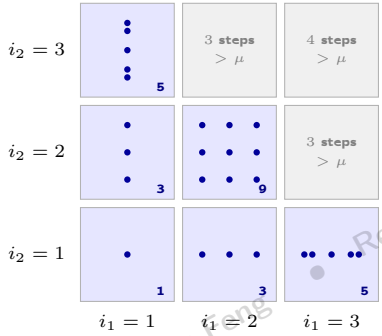
Not 5.B2's *zeros*, because zeros do not nest: the zeros of T_3 are $0, \pm 0.866$, and none but 0 is a zero of T_5. The extrema of level 3, $\{0, \pm 0.71, \pm 1\}$, contain all of level 2, $\{0, \pm 1\}$, so every value already paid for is reused. The *basis* is Chebyshev too: products of T_k, two frames on.

Malin, Krueger & Kubler (2011). Ledger `smolyak.construction` (4.B1's run); construction and the choice of μ: 4.B1.

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Worked Example: $d = 2$, $\mu = 2$, so $q = 4$

Each way to spend two steps is one small tensor grid; the sparse grid is their union.

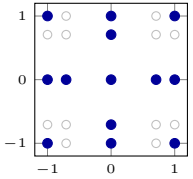


Cell (i_1, i_2) holds $G^{i_1} \times G^{i_2}$; the corner number counts its points. Union (right): **13 distinct points** of the 5×5 tensor grid's 25; the 12 hollow ones are dropped.

Ledger `fig_smolyak_2d`, `smolyak.index_sets_2d_mu2` **(4.B1's run)**.

Spend the budget. Each direction starts at level 1, free. The two steps go to x, $(3, 1)$; to y, $(1, 3)$; or one each, $(2, 2)$. Spending fewer gives $(1, 1)$, $(2, 1)$, $(1, 2)$: the six shaded cells, all of $i_1 + i_2 \leq q = d + \mu = 4$. The grey cells need 3 or 4 steps: fine in *both* directions at once.

Take the union. The cells hold $1 + 3 + 3 + 5 + 9 + 5 = 26$ points, but the levels nest, so they overlap:



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Smolyak Returns — as a Basis This Time

4.B1's grid said where to evaluate. The same levels now say which polynomials to keep.

In one direction, level i has m_i nodes, so it can carry degrees $0, \ldots, m_i - 1$. Each level **adds**:

level i	1	2	3	4
nodes m_i	1	3	5	9
adds	T_0	T_1, T_2	T_3, T_4	$T_5, \ldots, T_8$

For each admissible $\mathbf{i}$ ($|\mathbf{i}| \le q$), multiply what its levels add in each direction. The basis is all those products:

$$\hat{F}(x) = \sum_{\mathbf{k} \in \mathcal{K}(q,d)} \theta_{\mathbf{k}} \, T_{k_1}(x_1) \cdots T_{k_d}(x_d),$$

$\mathcal{K}(q,d)$ the degree vectors so obtained; $\theta_{\mathbf{k}}$ make a residual small (5.B4).

Example: $d = 2$

$\mu = 1$: $(1,1) \to 1$; $(2,1) \to T_1(x), T_2(x)$; $(1,2) \to T_1(y), T_2(y)$. **5 terms.**

$\mu = 2$ adds $(3,1) \to T_3(x), T_4(x)$; $(1,3) \to T_3(y), T_4(y)$; $(2,2) \to T_a(x)T_b(y)$, $a, b \in \{1,2\}$. $5 + 4 + 4 =$ **13 terms**.

Idea

Terms and nodes match level by level — 5 and 13, the next frame's counts. So Smolyak collocation uses the sparse set *twice*, as basis and as points: as many equations as unknowns.

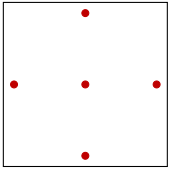
Smolyak (1963); in economics Krueger & Kubler (2004), Malin, Krueger & Kubler (2011), Judd, Maliar, Maliar & Valero (2014).

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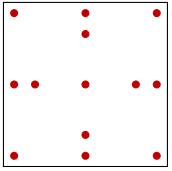
The Sparse Set in Two Dimensions, and What It Buys

Each level adds points along the axes first; off-axis points appear only when the index rule lets two directions refine at once.

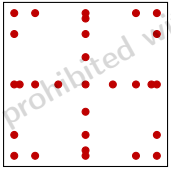
$\mu = 1$: 5



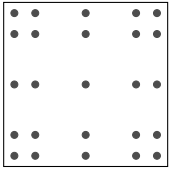
$\mu = 2$: 13



$\mu = 3$: 29



tensor 5×5: 25



d	Smolyak $\mu = 2$	tensor, 5 per dim
2	13	25
5	61	3,125
10	**221**	9,765,625

Weakened, not broken

For $f \in C^k$ the error bound is $C_d\, n^{-k} (\log n)^{(d-1)(k+1)}$: the one-dimensional rate spoiled by a logarithm whose exponent grows with d. Sparsity postpones the curse; it does not repeal it.

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Ledger `fig_smolyak_2d`, `smolyak` (4.B1's run). Bound: Barthelmann, Novak & Ritter (1999).

Smolyak Against Complete Polynomials

Complete polynomials cap the total *degree*; Smolyak caps the total *level*, and each level doubles the degree.

Same reach, $d = 2$**:** each keeps one variable up to degree 4 ($\kappa = 4$ against $\mu = 2$).

degree	complete	Smolyak
0–2	all 6	all 6
3	x^3, x^2y, xy^2, y^3	the same 4
4	$x^4, x^3y, x^2y^2, xy^3, y^4$	x^4, x^2y^2, y^4
count	**15**	**13**

Many variables, same reach:

d	complete, $\kappa = 4$	Smolyak, $\mu = 2$
5	126	61
10	1,001	**221**

Where the gap comes from

At $\mu = 2$ Smolyak keeps the constant, each variable alone to degree 4, and *pairs* to degree 2 each: $1 + 4d + 4\binom{d}{2}$ terms, never three variables at once. Complete keeps every product to total degree 4, e.g. $x_1x_2x_3x_4$.

The practical difference: nodes

Smolyak's levels bring a nested grid with as many points as terms: square collocation. Complete polynomials have no natural point set; Gaspar and Judd fit more points by least squares.

Counts: ledger `chebyshev.tensor_vs_complete` **and** `smolyak` **(the two previous frames). Gaspar & Judd (1997).**

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What a Term Costs, and What μ Allows

Steps are the currency, μ the budget, grid points what you pay for. A term is in if its price $\leq \mu$.

Step 1: a level carries a degree. m points pin down at most m coefficients, so a polynomial of degree $m-1$: three points fix a parabola, not a cubic.

level	nodes	up to degree	adds	steps
1	1	0	T_0	0
2	3	2	T_1, T_2	1
3	5	4	T_3, T_4	2
4	9	8	$T_5, \ldots, T_8$	3

Step 2: a term's price adds up across directions. $x^a y^b = T_a(x)T_b(y)$ needs the level that adds T_a in x and the one that adds T_b in y:

$$\text{price}(x^a y^b) = s(a) + s(b),$$

with $s(\cdot)$ the table's last column.

term	price	kept at $\mu = 2$?
x^4	$2+0=2$	yes
x^2y^2	$1+1=2$	yes
x^3y	$2+1=\mathbf{3}$	**no**
xy^3	$1+2=\mathbf{3}$	**no**

Where μ comes in

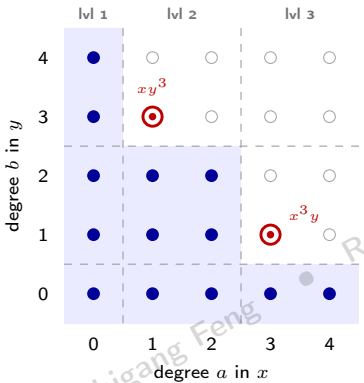
μ is the accuracy level you choose, like κ. Smolyak's rule $\sum_j (i_j - 1) \leq \mu$ *is* "price $\leq \mu$": $i_j - 1$ is the steps spent in direction j. At $\mu = 3$, x^3y comes in. Complete polynomials price by degree: x^3y and x^2y^2 both cost 4. Each step roughly doubles the points in its direction, so capping steps caps the grid.

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Malin, Krueger & Kubler (2011). Levels and nodes: 4.B1; the cells: the worked example above.

2 · Why x^3y and xy^3 Are Not in Smolyak

The price list, drawn — and why a price of 3 means the grid cannot fit the term.



● both (13) ◎ complete only (2) ○ tensor only

Malin, Krueger & Kubler (2011); the 13 terms are the previous frame's count.

Shaded: price $\leq \mu = 2$. Each block is one cell of the worked example; x^3y, xy^3 sit in $(3,2)$, $(2,3)$: price 3.

Why x^3y is out: the grid cannot see it

On the 13 grid points, $x^3y = T_3(x)T_1(y)$ takes *exactly* the values of xy: 0 on both axes, equal at the corners ($T_3(\pm 1) = \pm 1$). Two terms that agree at every point cannot both be fitted; Smolyak keeps xy.

Separating them needs points fine in x *and* off-axis in y at once: cell $(3,2)$, price 3, which $\mu = 2$ leaves out. x^2y^2 needs only the four corners, which the grid has.

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Finite Elements: The Opposite Choice

A spectral basis is global, so a kink anywhere spoils the fit everywhere. A finite-element basis is local: chop the domain, put a simple function on each piece.

Use cases: complex geometries or boundaries in the domain — in engineering, awkward shapes; in economics, a constraint.

Discretisation: partition the state space into small **elements** (intervals in 1-D; triangles or rectangles in 2-D). This **mesh** can adapt to curves and corners.

Basis functions: on each element a **piecewise polynomial**, equal to 1 at one node and 0 at all others — together a **partition of unity** (they sum to one).

Formulation and solution: substitute into the equilibrium conditions, integrate over each element, **assemble** the pieces into one algebraic system, and solve for the coefficients at the nodes.

Idea

Every spectral coefficient affects the answer everywhere. A finite-element coefficient affects it only near its own node. So a kink is not a problem — **it is a place to put an element boundary**.

The "irregular boundary" in macroeconomics is usually a constraint: $a' \geq -\underline{a}$, or $i_t \geq 0$. Put a node exactly there.

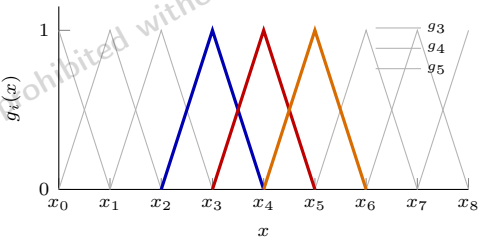
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The Simplest Basis: Tent Functions

Elements are small, so a linear basis is often good enough. Each tent is one at its own node, zero at every other, and the tents sum to one.

$$g_i(x) = \begin{cases} \dfrac{x - x_{i-1}}{x_i - x_{i-1}} & x \in [x_{i-1}, x_i] \quad \text{(rising)} \\ \dfrac{x_{i+1} - x}{x_{i+1} - x_i} & x \in [x_i, x_{i+1}] \quad \text{(falling)} \\ 0 & \text{elsewhere} \end{cases}$$

Then $d^n(x) = \sum_i \theta_i g_i(x)$, and θ_i is simply **the value of the policy at node i** — interpretable in a way a Chebyshev coefficient never is.



One-line check before solving: sum the tents on a fine grid. The correct basis gives exactly 1 everywhere; with the falling branch's sign flipped the sum is off by up to 2, and the solver still returns coefficients.

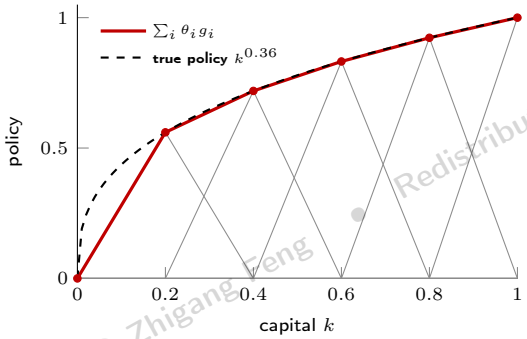
Nine nodes on $[0, 1]$. Each tent is nonzero only on $[x_{i-1}, x_{i+1}]$: it overlaps only its two neighbours.

Ledger `fem`: $\max_x |\sum_i g_i(x) - 1|$ **is** 0 **for the correct basis and** 2.0 **with the sign flipped.**

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A Policy Built From Tents

Scale each tent to the policy's value at its node and add them up: a piecewise-linear curve, exact at the nodes, cutting corners in between.



Six nodes on $[0, 1]$. Each thin tent is $\theta_i g_i(k)$, tent i scaled to the policy's value at node i. Their sum (red) is the **piecewise-linear interpolant**.

Idea: close up, a smooth curve is straight

Between two nearby nodes the chord stays close to the curve. On an element of width h the miss is at most $\frac{h^2}{8} \max |f''|$: halve h, quarter the miss. $k^{0.36}$ has $f'' \to \infty$ at 0, so the miss is largest there, and that is where smaller elements pay (two frames on).

Projection picks the θ_i by making a *residual* small, not from a known policy; the picture is the same.

Ledger `restored_figures.tent_weighted`. **Bound: linear-interpolation error, e.g. Atkinson (1989), ch. 3.**

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The Recipe, and Why the System Is Sparse

Plug the tent policy into the equilibrium conditions, one equation per node, and paste the pieces together. Each tent meets only its neighbours, so most entries are zero.

1: **partition** the state space into elements with nodes $x_0 < \cdots < x_n$
2: **basis:** a tent g_i at each node (a partition of unity)
3: **substitute** $d^n = \sum_i \theta_i g_i$ into the equilibrium conditions: $R(x;\theta)$
4: **for** $i = 0, \ldots, n$ **do** ▷ one equation per basis function
5: $\quad e_i \leftarrow \int g_i(x) R(x;\theta)\, dx$, element by element
6: **end for**
7: **assemble** the e_i into one system $A\theta = b$ and **solve**

Count: $n+1$ unknowns, $n+1$ equations. "Assemble" pastes neighbouring elements together at their shared node, so the result is continuous.

Why sparse

g_i is nonzero only on $[x_{i-1}, x_{i+1}]$, so $\int g_i g_j$ vanishes unless g_i and g_j share an element. Each row of A involves only neighbouring coefficients: the matrix is **banded**.

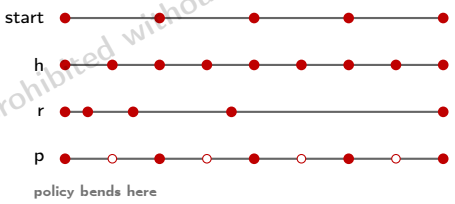
A banded system is solved in time proportional to n, a dense one to n^3: many more coefficients than Chebyshev, each far cheaper. Speed against accuracy decides.

Ledger `fem`: on nine nodes, 25 of 81 entries nonzero, bandwidth 1.

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More elements, better-placed elements, or higher order inside each element. Push the last far enough and the two families meet.

h-refinement	Subdivide each element into smaller ones: resolution improves uniformly. More nodes.
r-refinement	Keep the number of nodes; **move** them into the regions of high nonlinearity.
p-refinement	Raise the order of the approximation inside each element (hollow dots: extra nodes). High enough, and you have a hybrid of finite and spectral methods: **spectral elements**.



Where to put the elements

Small elements where the economy spends its time or the policy bends; a few large ones elsewhere. One element of degree n is a spectral method; n elements of degree 1 is FEM.

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Finite Elements Against Smolyak

They fail in opposite directions: dimension defeats finite elements, non-smoothness defeats Smolyak. Knowing which you face is the choice.

	Finite elements	**Smolyak**
Basis	piecewise polynomial, **local**; the solution may differ element by element	**global** Chebyshev polynomials (or splines) on a sparse index set
Grid	a **mesh**, which follows irregular boundaries and can be refined locally	tensor product of 1-D grids, thinned by the sparse rule
Strength	**low to moderate dimension** with awkward domains or kinks; widely used in engineering	**moderate to high dimension** with smooth solutions; far fewer points
Weakness	computationally heavy in high dimension: the mesh itself grows like n^d	less natural for irregular domains and non-smooth solutions

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Finite elements buy flexibility in **space** — where to refine. Smolyak buys economy in **dimension** — which interactions to keep. The Aiyagari economy of S6 has a kink and one asset: finite elements or a spline. A model with ten smooth states: Smolyak.

Takeaways

- **Part 1.** The tensor product multiplies one-dimensional polynomials together; orthogonality survives, but the basis has $(\kappa+1)^d$ terms — the curse of dimensionality again.
- **Part 2.** Complete polynomials cap the *total* degree, cutting $(\kappa+1)^d$ to $\binom{\kappa+d}{d}$ by dropping high-order interactions. Smolyak caps the total *level* instead: high degrees in one variable, little mixing, and its own square set of nodes — 221 terms at $d=10$ against $1{,}001$ complete and ten million tensor. The curse is postponed, not repealed.
- **Part 3.** Finite elements are the local basis: tents that are 1 at their own node and 0 elsewhere, so the coefficients *are* the policy values at the nodes, a kink becomes a node, and the system is banded. Refine by h, r or p.
- **Part 4.** Dimension defeats finite elements; non-smoothness defeats Smolyak.

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Bridge: What Does "Small" Mean?

Two decks have now been spent on the first of 5.B1's two questions — which basis. Spectral or local, Chebyshev or tents, dense or sparse.

The second question has been deferred every time it came up: having written $R(x;\theta)$, what exactly do we do to it? "Make it small" is not an algorithm, and the choices are not equivalent.

Next (5.B4)

Weighted residuals and the six answers: least squares, subdomain, moments, collocation, orthogonal collocation, Galerkin — tried on one model with one basis. Then the worked example: the same model a fourth way.

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